# CYBERGUARD SENTINEL

AI-Based User Behavior Anomaly Detection Platform

REPORT CLASSIFICATION: CONFIDENTIAL
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## 1. Executive Security Summary

This report summarizes behavioral telemetry, machine learning anomaly detection scores, and security rule evaluations across monitored user accounts. Anomalies are detected via Scikit-Learn Isolation Forest paired with behavioral baseline profiling and rule heuristics.

| Total Monitored Events | ML Anomaly Detections | High-Risk Incidents | Open Security Alerts |
|---|---|---|---|
| 41 | 5 | 5 | 8 |

## 2. Risk Score & Distribution Analysis

| Risk Tier | Score Range | Event Count | Proportion (%) | Operational Policy |
|---|---|---|---|---|
| LOW RISK | 0 - 29 | 33 | 80.5% | Standard logging; automatic baseline continuous learning |
| MEDIUM RISK | 30 - 69 | 3 | 7.3% | Elevated monitoring; baseline freeze; SOC analyst triage queue |
| HIGH RISK | 70 - 100 | 5 | 12.2% | Immediate alert raised; session termination / MFA step-up candidate |

## 3. High-Risk & Suspicious Activity Log (Recent 8 Incidents)

| Timestamp (UTC) | User | IP / Location | Target Resource | Risk Score | Flagged Detection Details |
|---|---|---|---|---|---|
| 2026-10-31 18:09 | admin | 127.0.0.1<br>Internal Portal Ingress | /dashboard | **33.7**<br>(Medium) | Connection initiated via penetration testing OS or anonymized client: unknown os / unknown browser. |
| 2026-10-31 18:09 | alice.chen | 45.154.255.89<br>External Suspicious Ingress | /api/v1/auth/login | **100.0**<br>(High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.77).; Login attempt at unusual hour (04:00) deviates from established user schedule. |
| 2026-10-31 18:09 | david.kim | 103.152.220.10<br>External Suspicious Ingress | /financial/payroll | **93.7**<br>(High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.62).; Login attempt at unusual hour (23:00) deviates from established user schedule. |
| 2026-10-31 18:09 | eva.rossi | 91.240.118.23<br>External Suspicious Ingress | /database/credentials | **100.0**<br>(High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.74).; Login attempt at unusual hour (01:00) deviates from established user schedule. |
| 2026-10-31 18:09 | bob.martinez | 194.26.29.112<br>External Suspicious Ingress | /admin/system-logs | **100.0**<br>(High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.71).; Login attempt at unusual hour (03:00) deviates from established user schedule. |

| | | | | | |
|---|---|---|---|---|---|
| 2026-10-31 18:09 | alice.chen | 185.220.101.5<br>External Suspicious Ingress | /api/v1/auth/login | **100.0** (High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.79).; Login attempt at unusual hour (02:00) deviates from established user schedule. |
| 2026-10-31 18:09 | eva.rossi | 10.0.12.25<br>Corporate HQ / Domestic | /projects/tickets | **31.5** (Medium) | Login attempt at unusual hour (06:00) deviates from established user schedule. |
| 2026-10-31 18:09 | alice.chen | 192.168.1.45<br>Corporate HQ / Domestic | /documents/shared | **37.1** (Medium) | Login attempt at unusual hour (06:00) deviates from established user schedule. |

## 4. Recommended Security Actions & Remediation

**1. Multi-Factor Authentication (MFA) Step-Up:** Enforce mandatory FIDO2 / TOTP challenge for logins originating from new IP addresses or unfamiliar client operating systems.

**2. Account Lockout & Throttling:** Configure automated temporary lockout or exponential rate-limiting for accounts accumulating 3+ failed logins in 5 minutes.

**3. Baseline Safeguard Enforcement:** Keep behavioral baseline profiles locked for users undergoing internal investigation to prevent adversarial profile poisoning.

**4. Geofencing & Off-Hours Ingress Policies:** Implement conditional access policies restricting administrative asset access between 23:00 and 06:00 UTC without prior SOC clearance.

**5. SOC Incident Response:** Review all OPEN High and Critical security alerts within a 15-minute SLA. Resolve alerts with thorough forensic notes.